A data-Driven Technique For Workforce Productivity Prediction in The Garment Industry Using Supervised and Unsupervised Machine Learning Models with Explainable AI

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***Abstract*—The garment industry is a key component of global economic activity, employing millions of workers worldwide and significantly impacting regions such as East Africa. However, it faces a persistent challenge: the productivity gap between actual and targeted employee performance, which results in production delays and financial losses. Addressing this issue is crucial for improving operational efficiency, enhancing global competitiveness, and fostering economic development. This study investigates the application of machine learning (ML) techniques to predict employee productivity within the garment industry. Through exploratory data analysis, the study identifies key variables that influence productivity. Various machine learning algorithms—including Multi-Linear Regression, Random Forest, Lasso Regression, and XGBoost Regression—are experimentally applied, with model performance evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Square Error (RMSE), and R^2 score to compare their predictive accuracy. The results indicate that the random forest regression model achieved the lowest MAE value of 0.0761. As for MSE value, the XGBoost Regression model had the lowest value at 0.0191, followed closely by the Random Forest model with a value of 0.0196. Additionally, the study incorporates an explainable AI technique called LIME to interpret how the models arrive at their decisions. through the application of explainable AI using LIME,the best performing models were Random Forest model and the XGBoost Regression model. The findings underscore the potential of machine learning-driven approaches coupled with explainability to enhance employee management and promote sustainable practices, particularly in the garment industry in regions like East Africa.**

***Index Terms*—Machine learning, Explainable Artificial Intelligence (XAI), Exploratory Data Analysis (EDA), Productivity prediction.**

I. Introduction

The garment industry is one of the most labor-intensive industries in the world. About 75 million people are directly involved in the textile, clothing and footwear sector worldwide.(united, 2021) [1]. In Uganda, the market is projected to experience a volume growth of 3.7 percent in 2025 [2]. A common problem in this industry is that the actual productivity of apparel employees fails to reach the productivity targets set by the authorities to meet production targets on time, resulting in huge losses. With the industry's labor-intensive processes and need to meet the ever-growing global demand for garments, tracking and enhancing the productivity performance of employees has become a top priority for decision-makers in the industry. Before increasing the productivity of employee performance, it is necessary to know in advance what factors affect and how to predict employee productivity, especially garment employees that are being discussed. This analysis aims to solve the productivity gap problem in the garment industry by predicting the actual productivity of employees.

II. Background and motivation

The garment industry, like any other labor-intensive industry, faces the common issue of employee productivity not meeting the actual productivity set by management, leading to significant losses. Employees in workplaces usually have different working aptitudes and paces and because of these, industries often fail to meet the target productivity required by the authorities to achieve production goals over a given period of time. To solve this problem of loss of achieving the target productivity outcome, there are some techniques that companies utilize for example systematic monitoring and analysis of production level by keeping track of overall performances of teams. This can be tedious to do manually since for the garment industry there are many individuals working in different departments and teams as explained in [3]. Multiple regression and classification models to help in predicting employee productivity were explored and used in this paper. With this machine learning based approach, the hustle of manual productivity analysis and tracking can be minimized thus allowing companies to make better decisions.

Machine learning;this is a subset of Artificial intelligence that allows computers to learn and improve without explicitly being programmed, machine learning systems use algorithms and statistics to analyze erroneous amounts of data, identify patterns and make predictions. Machine learning models are split into four categories [4]. These include supervised machine learning which is defined by its use of labeled datasets to train algorithms to classify data or predict outcomes accurately, Unsupervised Machine learning uses machine learning algorithms to analyze and cluster unlabeled datasets. Semi-supervised machine learning utilized both supervised and unsupervised learning. During training, it uses a smaller labeled data set to guide classification and feature extraction from a larger and unlabeled data set. Reinforcement machine learning is a model that learns as it goes by using trial and error. A sequence of successful outcomes will be reinforced to develop the best recommendation for a given problem. Some of the commonly used machine learning algorithms include Linear regression, Random forests, Logic regression, Neural networks, Clustering and decision trees.

Explainable Artificial intelligence (XAI); this refers to a set of processes and methods that allows humans to comprehend and trust the results and outputs generated by machine learning algorithms [5]. Explainable AI is important to the users who utilize the AI system. When the AI recommends a decision, the decision makers would need to understand the underlying reason. Explainable AI could help developers to improve AI algorithms, by detecting data bias, discovering mistakes in the models, and remedying the weakness [6]. Its goal is to develop a set of techniques that either produce more understandable models keeping high levels of performance or provide external tools to better understand the models that are inherently not interpretable [4]. The methods used to interpret machine learning models are categorized into two, Model-agonistic methods and model-specific methods. Model-agonistic methods Include Local Interpretable Model-agnostic Explanations (LIME) and SHapley Additive exPlanations (SHAP). LIME is an algorithm that focuses on explaining the model's prediction for individual instances. LIME can explain individual predictions of any classifier or regressor in a faithful way. SHAP aims to explain the prediction of an instance by computing the contribution of each feature to the prediction [5] [7]. For this study LIME was applied to understand and explain the models' predictions.

Exploratory data analysis(EDA); this is an approach of data analysis that employs a variety of techniques, most especially graphical in order to maximize insights into a dataset, uncover underlying structure, extract important variables, detect outliers and anomalies, determine optimal factor setting among others. it also provides tools for hypothesis generation by visualizing and understanding the data usually through graphical representations [8]. EDA can be classified as graphical or non graphical methods, univariate or multivariate methods.

Productivity prediction refers to the process of anticipating the future output, performance or even efficiency of an individual, team or system based on different inputs and variables. Productivity prediction involves analyzing historical data, current trends and potential factors to estimate how productive a person or entity will be in a given period of time or under certain conditions. Productivity refers to the measure of efficiency that compares the amount of output produced to the amount of inputs used in a given process [9]. It is a relative concept, emphasizing how effectively resources like labour, capital, and materials are utilized. Factors that affect productivity include compensation, work environment, training, career development among others. Productivity can be demonstrated at work through setting goals, focusing on one task at a time, meeting deadlines among others. In the garment industry, productivity is influenced by several factors categorized into controllable (internal) and uncontrollable (external) factors Atsegeba et al [10]. These factors include Work-in-progress (WIP) accumulation, throughput time, balancing the distribution of tasks and workloads, transportation and space utilization, quality control among others.

III. LITERATURE REVIEW

Ensemble learning algorithms such as decision tree, AdaBoost, Naïve Bayes, Random Forest and SVM were applied by [11] for predicting employee productivity. The results proved that the machine learning models predicted productivity of employees with a high accuracy. Integration of Physiological and behavioral features with machine learning techniques such as random forest XGboost, logic regression were used by Muhammad et al 2023 [12] to predict productivity of office workers and these models equally proved that certain physiological features greatly affect productivity such as stress. Similarly, physical, social, and economic environmental factors along with classification machine learning algorithms such as Logistic Regression classifier, the Gaussian Naive Bayes, the Decision Tree classifier, the K-Nearest Neighbors (K-NN) among others were used by Zannatuf and Aftab 2023 [13] to provide an unbiased artificial intelligence algorithmic solution to predict future employee performance evaluation for better decision making.

A. Research gaps in the literature

Even with significant progress in predictive models concerning employee productivity, there are still notable deficiencies present, including the following:

- Many proposed models are black-box models, which lack explainability and interpretability. This makes it difficult for managers to trust and understand the decisions made by these models.
- There is limited research focusing on region-specific factors, particularly in areas like East Africa. This shortcoming is largely due to the unavailability or inaccessibility of data from local industries.
- Most existing models primarily explore supervised learning techniques for predicting employee productivity. While these techniques can be effective, they may overlook valuable insights that could emerge from unsupervised or semi-supervised approaches,

B. contributions

This paper presents a comprehensive model designed to effectively predict employee productivity and significantly enhance our understanding of workforce dynamics specifically in the garment industry. The key contributions of this paper are as follows;

- We suggest a sophisticated regression approach utilizing machine learning techniques to accurately forecast the productivity levels of employees. This method harnesses various features related to employee performance and workplace dynamics, allowing for an in-depth analysis and reliable predictions.
- In addition to regression analysis, the study includes an exploration of unsupervised learning techniques. This approach allows us to identify hidden patterns within dataset clusters providing insights into underlying employee productivity drivers without predefined labels.
- Explainable and interpretable artificial intelligence. By incorporating techniques that improve the interpretability of machine learning output, the study aims to build trust among stakeholders with respect to the predictions of the model. Specifically, the research employs LIME (Local Interpretable Model-agnostic Explanations) to provide transparent insights into how different input features influence the model's decisions, thereby addressing the common concerns associated with black-box algorithms.

Through these contributions, our goal is to improve the predictive accuracy of employee productivity assessments while addressing the need for model transparency in practical applications.

IV. Methodology

The dataset being used consists of 1197 instances accessed as a CSV file and loaded into python panda's data-frame for further prepossessing. The data prepossessing procedure included identifying and handling missing values, feature selection,outlier detection and removal, feature encoding, feature scalar and then data partitioning. Missing values were found in column wip that is to say 506 missing values that contributes to 42.273 percent of the total data in the column, these were replaced with 0. Columns incentive, wip and over-time had the most significant outliers and these were handled by interquartile range and capping. Miss-spellings were present in the department column and corrected. Data visualization using bar plots, line graphs, box plots heat maps were used to visualize the data and relationship between the columns. After the prepossessing, feature encoding using onehot encoder was applied to the categorical columns such as quarter, day, etc. The standard scaler was applied to normalize the data and for the data splitting, 80 20 ratio was used with 80 percent for training and 20 percent for testing data. Machine learning models including Random forest, lasso regression, Multilinear regression and XGBoost were experimented applying hyperparamter tuning and feature importance analysis for each model. Explainable AI technique applied was lime which was applied on each of the four models testing on three instances each. The results are explained in the next section. As part of unsupervised learning, k-means clustering and the specified machine learning models were experimented on the clusters. there is a simple flow diagram to illustrate the methodology.

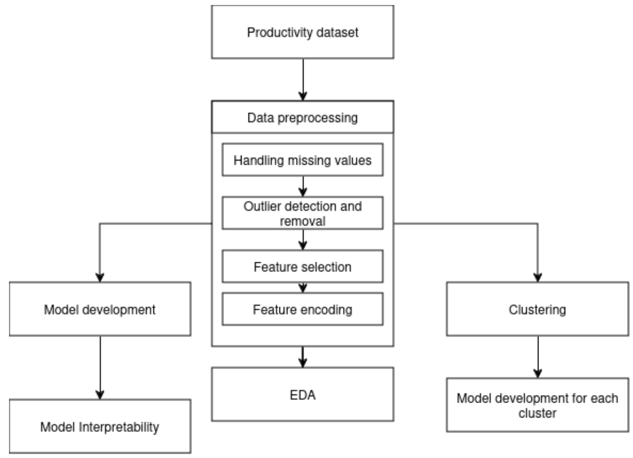

Fig. 1. Methodology flow diagram.

A. Dataset description

The dataset was sourced from a renowned and reputed garment industry in Bangladesh, which is presented by the Industrial Engineering department of the company and made publicly available through UCI Machine Learning repository in 2020 [14]. The data was collected for three months starting from January 2015 to the end of March 2015. It contains the production data of the sewing and the finishing team out of many other teams in the industry. It particularly has 1197 data instances as well as 13 attributes. This dataset was selected for this study because it is important for model development to address the problem of accurate employee productivity prediction, the following is a brief summary of the considerations when selecting the dataset.

- A labor intensive industry since we're dealing with employee productivity prediction.
- List of features. The features captured in the dataset are generic, widely applied and are crucial in performance evaluation even for garment industries locally in East Africa for example incentive, over-time, target productivity among others.
- Dataset availability and quantity for experimentation. The dataset is free and available in a public repository and consists of fairly enough data to work with.

The attributes are highlighted below with their functions in the dataset.

1) Date: It is mainly the date of the month mentioned in the particular year in the 'month-date-year' format.

2) Department: It is an attribute that states the associated department along with the instance.

3) Team_no: Numbered team that is associated with the instance.

4) No_of_workers: Numeral for workers, a specific team holds.

5) No_of_style change: Changes made in quantity, in a particular styled product.

6) Targeted productivity: The target to fulfill for each team, which is set by the authorities.

7) SMV: Standard Minute Value. (Amount of time fixed for a certain task).

8) WIP: Work in Progress. (Quantity of unfinished items in a garment).

9) Over time: Minutes of overtime done by a team.

10) Incentive: Financial incentives enabling a particular task.

11) Idle_time: Time that was idle or unproductive due to several kinds of production interruptions.

12) Idle_men: Workers who were idle during the production interruption period.

13) Actual_productivity: Productivity value of the workers on the scale of the expected productivity, ranging from 0.0 to 0.1.

B. Data Preparation and Exploratory Data Analysis

During the preprocessing phase some of the columns were reviewed and shown to have missing values, outliers and miss-spellings which all had to be dealt with in the following ways.

1) *Handling missing values*. Checking for null values in the data frame showed that the column wip (work in progress) had some missing values constituting 42.27 percent of the total amount of data in the column. These null values were replaced with zero because through observation the value of wip at the finishing department is supposed to be zero.
2) *Outlier detection and removal*. Application of outlier detection and removal is crucial in machine learning as outliers may skew the results of the machine learning algorithms. Through the use of box plots outliers were detected in the 'incentive', 'wip' (Work in progress) and 'over_time' columns as shown in figure 2. The removal of outliers was done through interquartile range and capping. This involved finding the first and third quartiles then determining the interquartile range or IQR. values for the lower limit and upper limit of the data were calculated and then used to remove the values in between to remove all the outliers. The data frame after removal of outliers is shown in figure 3.

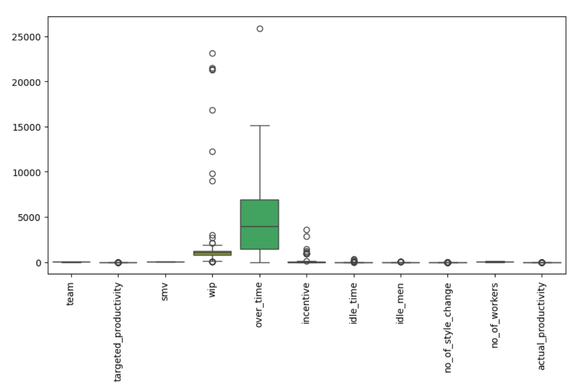

Fig. 2. Box plot showing of the columns before outlier removal.

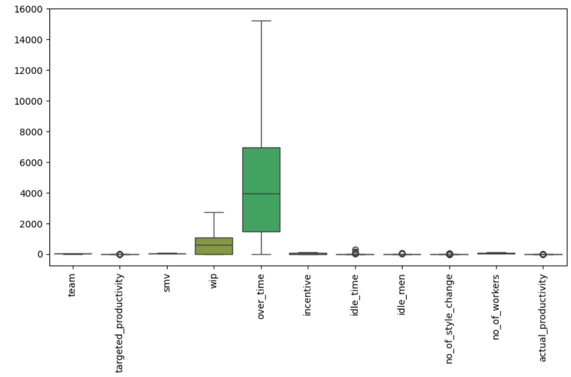

Fig. 3. Box plot showing of the columns after outlier removal.

3) *Feature encoding*. This is the process in which categorical variables are characterized into numericals. This is because machine learning models learn strictly from numeric data. Columns such as department', 'Team_no', 'month', 'quarter' were encoded into numerical data using label encoding.
4) *Exploratory data analysis*. this involves in depth analysis of the dataset using different graphs to visualize the relationship between the columns and their overall distribution.
   - Figure 4 displays a pair plot for each column of the dataset.This helps to get a quick overview of the relationships between data variables and their distribution.

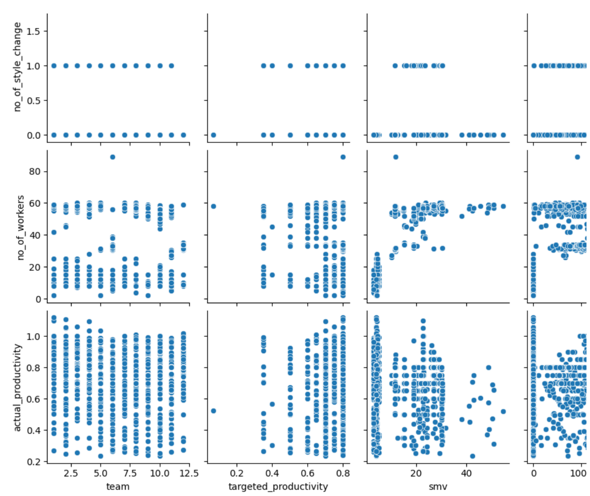

Fig. 4. Pair plot showing for all columns columns.

   - Figure 5 shows a correlation heat map used for depicting the correlation between columns in the dataset. It helps visualize the strength and direction of linear relations between variables. For example There is a significantly high positive correlation between 'no_of_workers' and 'smv' (Standard Minute value) and a weak correlation between 'over-time' and 'smv'(standard minute value).

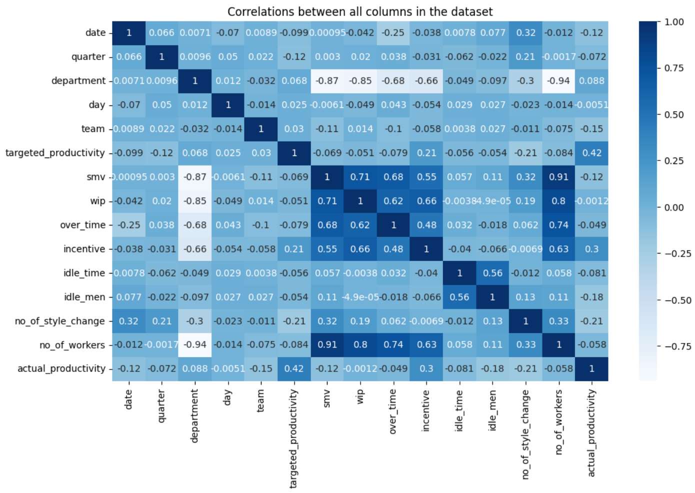

Fig. 5. Heat map showing correlations between columns.

- Based on figure 6 it can be inferred that the sewing department contributed more to the garment productivity compared to the finishing department. This is because the slope of the sewing department has a stronger linear relationship compared to the finishing department.

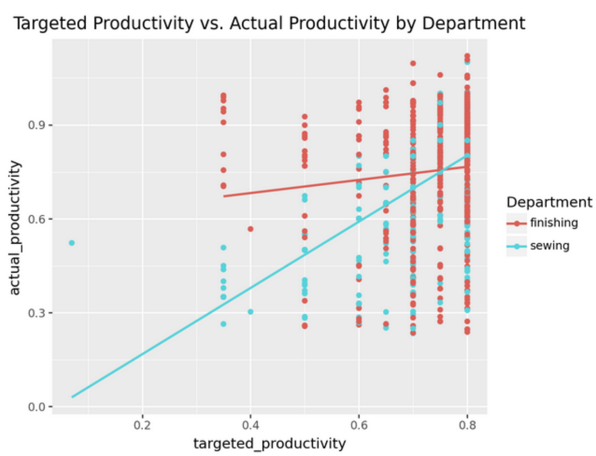

Fig. 6. Scatter plot for actual productivity plotted against target productivity by department.

- *Onehot encoding*. One Hot Encoding is a method for converting categorical variables into a binary format. It creates new binary columns (0s and 1s) for each category in the original variable. This resulted in more columns for example department_sewing, department_finishing, day_monday, day_tuesday to name a few.
- *Dependent and independent column selection*. After Onehot encoding, the x metrics(feature columns) and y metric (target column) are selected.
- *Data splitting*. This involves dividing the data into two or more subsets, random split was performed using scikit-learn train-test-split library the data was divided into a training set (x-metrics-train,y-metrics-train) and a testing set (x-metrics-test, y-metrics-test), allocating 20 percent of the data to the testing set and 80 percent to the training set.
- *Feature scalar*. This is a technique used in data standardization which scales the data within a fixed range. The standard scaler was applied by fitting the training set and then transforming the test set. The reason for this was to prevent data leakage caused by fitting an entire dataset.

C. Machine learning model selection and optimization

Four machine learning models were used for this study including Multi-linear regression, Random forest regression, lasso regression and XGBoost regression. GrideSearch cross validation, a technique used for hyper parameter tuning was used for better results of the models. The models and hyper-parameters are explained below.

1. *Multi Linear regression model*. This is a statistical technique that uses a straight line to estimate the relationship between a dependent variable and multiple independent variables [15]. Multi-linear regression model is used to predict the outcome of a variable based on the values of other variables. It relies on several key assumptions to produce valid and reliable results namely Linearity, independence, Homoscedasticit among others. It finds the best line that minimizes the differences between predicted and actual values. Parameter used

- Fit_intercept, boolean, default is True. it determines whether to calculate the intercept for this model or not depending on the boolean variable given.

2. *Random Forest regression model:* this is an ensemble model that combines the predictions of multiple decision trees to reduce over-fitting and improve accuracy[] .The approach's significant capacity for learning, resilience, and feasibility of the hypothesis space account for the benefits. In a random forest, each node is split using the best among a subset of predictors randomly chosen at that node. This somewhat counterintuitive strategy turns out to perform very well compared to many other classifiers [16]. Parameters used include;

- N_estimators. Number of trees the algorithm builds before averaging the predictions
- Max-depth. The maximum depth of the decision trees. Deeper trees can capture more complex patterns in the data,
- min_sample_split, The minimum number of samples required to split a node. A larger value can prevent over-fitting, but may also make the model less flexible.
- Min_samples_leaf. The minimum number of samples required to be at a leaf node Random state, a fixed random state used to control any such randomness involved in machine learning in order to get consistent results.

Hyper-parameters values obtained include n_estimators - 200, min_samples_split': 10, min_samples_leaf': 4

3. *Lasso regression model*.This a machine learning regression model that employs L1 regularization its an extension of the linear regression model, which adds a penalty equal to the absolute value of the coefficients. This characteristic allows Lasso to set some coefficients exactly to zero. The lasso model is an effective strategy to handle multicollinearity and prevent overfitting [17]. It is used when there are many features because it performs feature selection automatically. Parameters include Alpha, this is the regularization parameter. Hyper-parameter value obtained Alpha: 0.001

4.*XGBoost regression model*. XGBoost is a decision tree ensemble based on gradient boosting designed to be highly scalable. Similarly to gradient boosting, XGBoost builds an additive expansion of the objective function by minimizing a loss function [18]. Considering that XGBoost is focused only on decision trees as base classifiers, a variation of the loss function is used to control the complexity of the trees. Parameters

- Max-depth, The maximum depth of a tree. Used to control over-fitting.
- Max-leaf-nodes, The maximum number of terminal nodes or leaves in a tree.
- Min_child_weight, Defines the minimum sum of weights of observations required in a child.
- Learning-rate or eta. is a parameter that controls the rate at which the algorithm learns from each iteration.

Hyper-parameter values obtained include Learning-rate or eta: 0.01, Max-depth: 5 and N-estimators: 300

D. *Machine learning model selection Accountability*

Explainable AI is a technique used to describe an AI model, its expected impact and potential biases examples include LIME and SHAP [5]. This aids in characterization of model accuracy, fairness, transparency and outcomes in an AI powered system of decision making. Explainable AI is important for an organization to build the trust and confidence required when deploying AI models in real world scenarios. AI accountability relates to the assurance that AI is functioning properly throughout its entire life-cycle. A key component of AI accountability is the ability to interpret and understand what the AI is doing and how it arrives at a particular conclusion without treating it as a Black-Box. The explainable AI technique used for this study is LIME which was applied to all the models to get an understanding of how the models arrive at their outcomes. For more accuracy multiple instances must be tested to arrive at a conclusion. For this case, instance 1, 20 and 50 were used for each of the four models used in the study.

Figure 7 shows LIME graphical representation for the best model that was the random forest. The outcome is similar to the feature importance graph plotted for the random forest model as shown in figure 8.

The explainability of this model indicates that the factors contributing positively to productivity include target productivity with a value of 0.13, incentive with a value of 0.10, the number of workers with a value of 0.30, and department with a value of 0.13. This rationale is understandable because targets motivate employees to achieve or even exceed their productivity goals. Additionally, the number of workers can enhance productivity; a larger workforce often allows for more work to be completed, depending on the job. Furthermore, some departments are inherently more demanding than others, which can impact employee productivity.

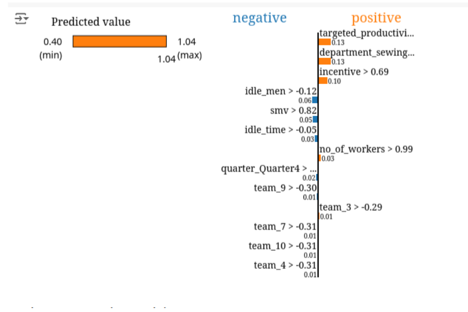

Fig. 7. LIME representations of the Random forest interpretation on instance 20.

Feature	Value
targeted_productivity	0.80
department_sewing	1.00
incentive	38.00
idle_men	0.00
smv	25.90
idle_time	0.00
no_of_workers	56.00
quarter_Quarter4	0.00
team_9	0.00
team_3	0.00
team_7	1.00

Fig. 8. LIME representations of the Random forest interpretation on instance 20.

On the other hand, the negative contributors to productivity include idle men with a value of 0.06, standard minute value (SMV) with a value of 0.05, idle time with a value of 0.03, quarter 4 with a value of 0.02, and team 9 with a value of 0.01. These factors have the highest negative impact on the productivity prediction, although their percentages are relatively small.

Department 1 (sewing department) likely influenced the

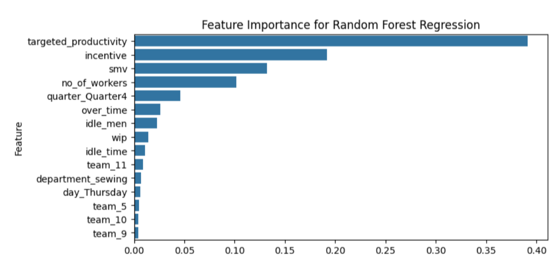

Fig. 9. Random forest feature importance.

productivity prediction more significantly than Department 2 (finishing) because it has higher performance records. Incentives act as a direct motivator for employees to increase effort and achieve better outcomes the higher the incenitve the better the productivity.

E. Results and Discussion

Evaluation metrics

The proposed model is assessed using four key metrics referenced in [19]: Mean Squared Error (MSE), Root Mean Square Error (RMSE), R-squared (R^2) score, and Mean Absolute Error (MAE).

1. Mean Squared Error (MSE) evaluates the average of the squares of the errors, providing an understanding of how close a model's predictions are to the actual outcomes. A lower MSE indicates a better model fit. This is calculated using the equation

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

2. Root Mean Square Error (RMSE) is the square root of the MSE and provides insight into the typical magnitude of the prediction errors in the same units as the original data. This metric is particularly useful for understanding the average error size and is calculated using the equation

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

3. R-squared (R^2) score measures the proportion of variance in the dependent variable that can be predicted from the independent variables. This statistic provides insight into how well the model explains the variability of the data, with values closer to 1 indicating a better explanatory power. Below is the equation to calculate R2 score

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

4. Mean Absolute Error (MAE) calculates the average absolute differences between predicted and actual values, offering an intuitive measure of prediction accuracy. MAE is beneficial for understanding the average magnitude of errors in a straightforward manner, without regard for their direction.MAE is also referred to as a loss function since the goal is to minimize this loss function and is defined as follows

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

By employing these comprehensive metrics, we can gain valuable insights into the performance and reliability of the proposed model.

Results

Analysis of the results obtained from the machine learning algorithm was done using the above specified evaluation metrics. The values taken to represent the results are record in table 1. These values were rounded off to 4 decimal places. We input the summarized values that were obtained from the machine learning models after hyperparameter tuning and state them in the table 1. We wound the lowest MAE value of '0.0761' for the random forest model and the highest value of '0.1083' from the Linear regression model. The lowest RMSE values of '0.1383' for XGBoost regression model and the highest value of '0.1542' from lasso regression model. A MSE of 0.0191 was the lowest obtained from XGBoost and the highest value '0.0238' from lasso regression model. The highest R2 score value of '0.3730' was obtained from the XGBoost regression model and the lowest R2 score value of '0.2211' obtained from the lasso regression model. Table of results

Models	**MAE**	**RMSE**	**MSE**	**R^2 Score**
Linear Regression	0.1083	0.1522	0.0232	0.2409
Lasso Regression	0.0966	0.1542	0.0238	0.2211
Random Forest Regression	0.0761	0.1400	0.0196	0.3582
XGBoost Regression	0.0800	0.1383	0.0191	0.3730

TABLE I

COMPARISON OF EVALUATION METRICS FOR REGRESSION MODELS

From the above table of results it is evident that the random forest model and XGBoost model outperformed the other models for all the evaluation metrics. For this study we consider MAE as the main evaluation metric because It gives equal weight to all errors and the main goal is to minimize the overall error in the model while avoiding large errors. The lesser the MAE value the better the model. The random forest regression model gave the lowest MAE value therefore we consider it as the best model in accordance with our findings.

The results in the table above clearly indicate that both the random forest model and the XGBoost model outperformed the other models across all evaluation metrics. In this study, we have chosen Mean Absolute Error (MAE) as the primary evaluation metric because it treats all errors equally, and our main objective is to minimize the overall error in the model while avoiding significant errors. A lower MAE value indicates a better model. Given that the random forest regression model produced the lowest MAE value, we consider it the best model based on our findings.

F. Unsupervised learning using clustering

As for unsupervised learning. We applied Kmeans clustering and obtained 2 as the optimal number of clusters using the Elbow method. To improve segmentation results Principal Component Analysis (PCA) was appllied on the data. Principal Component Analysis (PCA) is an unsupervised learning algorithm technique used to examine the interrelations among a set of variables. The main goal of Principal Component Analysis (PCA) is to reduce the dimensionality of a dataset while preserving the most important patterns or relationships between the variables without any prior knowledge of the target variables. Figure 10 shows a visualization of pca on the dataset with the 2 clusters obtained. Completeness score which measures how well the data points of a particular class are assigned to the same cluster and homogeneity score which measures how similar the data points are within a cluster were used to evaluate the quality of a clustering model and they both yielded 1 meaning the clustering was perfect.

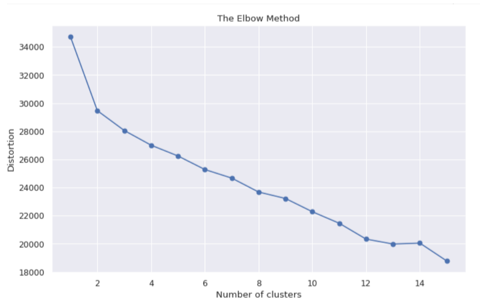

Fig. 10. Kmeans clustering plot with two dimensions.

The Four machine learning models specified above namely Linear regression model, lasso regression mode, Random forest regression model and XGBoost regression model were applied to each of the clusters and the specified evaluation metrics namely MAE, MSE, RMSE and R2 score were used to evaluate each of the models for each of the 4 clusters. Once more considering MAE as the main evaluation metric, the table 2 shows a summary of results of MAE value from each of the clusters for each of the models

Clusters	**Linear Reg**	**Random Forest**	**Lasso**	**XGBoost**
Cluster 0	0.136808	0.232902	0.111500	0.357502
Cluster 1	0.0736697	0.811142	0.741899	0.808361

TABLE II
MAE VALUES FOR CLUSTERS ACROSS DIFFERENT MODELS

G. Conclusion and future works

The aim of this study was to propose a machine learning model to be used for evaluating productivity of employees in the garment industry. The database was collected from a big and well known garment production company in Bangladesh. Preprocessing of the data was carried out using sequential procedures such as missing value detection and handling, outlier detection and handling, feature encoding, feature scaling and data splitting. Four machine learning models were used namely LInear regression model, lasso regression mode, Random forest regression model and XGBoost regression model and evaluation metrics namely MAE, MSE, RMSE and R2 score were applied to each of the models. The random forest model proved to be the best model for predicting productivity of employees in both supervised and unsupervised approaches in terms of the MAE evaluation value. Explainability using Lime also shows that the random forest is the best model considering the following features positively influencing productivity prediction; target productivity, incentive, no-of-workers and department. This highlights the importance of having a target which always motivates employees to work hard and reach the target, incentive is another factor that highly motivates employees because with little or no incentive demotivates employees. The number of workers and departments are equally important and contribute to the productivity of employees in the garment industry. As part of future works, we would like to investigate further the low value obtained from calculating the R2 score which represents the proportion of the variance for a dependent variable that's explained by an independent variable because the best performing model in terms which is XGboost in terms of R2 score give a value of 0.37 which is below average. Investigate and apply machine learning models for productivity prediction of garment employees to locally obtained datasets from east africa so that the region can benefit from the study and finally deploying the model.

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